Syllabus Details

Syllabus ID:	13778
Syllabus Name:	**Advanced Game Development_Phát triển game nâng cao**
Course Name English:	**Advanced Game Development**
Subject Code:	**AGU301**
Learning-Teaching Method:	Project-Based Learning, In-class lecture, Active Learning
NoCredit:	3
Degree Level:	Bachelor
Time Allocation:	45h (60 sessions) contact hours + 1h final exam + 104h self-study
Pre-Requisite:	FGU301
Description:	This advanced Unity course builds on foundational skills to empower you with expert-level knowledge in creating professional, high-performance games. You'll dive deep into SOLID principles and design patterns for writing clean, scalable code, while mastering Unity's networking tools and cloud services to build engaging multiplayer experiences. Hands-on projects will help you optimize performance, integrate AI, and deliver polished games ready for consoles, PC, and beyond. Key Learning Outcomes: - Master SOLID Principles: Build a solid grasp of the five SOLID principles (Single Responsibility, Open-Closed, Liskov Substitution, Interface Segregation, and Dependency Inversion) and apply them to make your Unity code more modular and easier to maintain. - Implement Design Patterns: Get hands-on with essential Unity patterns like Singleton, Observer, Command, Strategy, Factory, Decorator, and State to organize code, boost flexibility, and simplify complex game logic. - Build with Unity Networking: Explore Unity Relay and Netcode to design smooth, reliable multiplayer systems that handle real-time interactions effortlessly. - Integrate Unity Gaming Services: Use cloud tools like Player Authentication, Lobbies, In-Game Economy, and Leaderboards to add secure, scalable features that enhance player engagement. - Optimize Performance: Discover profiling techniques and best practices to fine-tune your games for buttery-smooth gameplay, reduced lag, and efficient resource use on various devices. - Create Multiplayer Games: Develop small real-time multiplayer prototypes using Unity's networking and collaboration tools, following industry-standard workflows. - Leverage Unity AI: Incorporate AI models via Unity Sentis or similar tools to add intelligent behaviors, like smart NPCs or procedural content, to your game projects. By the end of this course, you'll have the confidence and skills to solve tough game development problems, from code architecture to live multiplayer features, and produce professional-quality games with cutting-edge design and tech.
StudentTasks:	- Students must attend more than 80% of contact slots in order to be accepted to the final examination. - Student is responsible to do all exercises, assignments and labs given by instructor in class or at home and submit on time - Use laptop in class only for learning purpose - Promptly access to the FU CMS at https://lms-hcmuni.fpt.edu.vn/ for up-to-date course information
Tools:	
Scoring Scale:	10
DecisionNo MM/dd/yyyy:	1318/QĐ-ĐHFPT dated 11/27/2025
IsApproved:	**True**
Note:	
Is Scored:	**True**
MinAvgMarkToPass:	5
IsActive:	True
ApprovedDate:	11/27/2025

7 material(s)

No.	Material Description	Author	Publisher	Published Date	Edition	ISBN	Is Main Material	Is Hard Copy	Is Online	Note
1	Unity 6.2 User Manual.	Unity Technologies	Unity Technologies				True	False	True	https://docs.unity3d.com/Manual/index.html
2	A tour of the C# language	Microsoft	Microsoft				True	False	True	https://learn.microsoft.com/en-us/dotnet/csharp/tour-of-csharp/overview
3	Unity's official e-book: Level up your code with design patterns and SOLID	Unity Technologies	Unity Technologies	2024			False	False	True	https://unity.com/resources/design-patterns-solid-ebook
4	Unity's official e-book: Optimize your console and PC game performance	Unity Resources	Unity Technologies	2023			False	False	True	https://unity.com/resources/optimize-your-console-and-pc-game-performance
5	Unity's official e-book: Ultimate Guide to Profiling Unity Games	Unity Resources	Unity Technologies	2023			False	False	True	https://unity.com/resources/ultimate-guide-to-profiling-unity-games
6	Unity's official e-book: The ultimate guide to multiplayer networking for advanced Unity developers	Unity Resources	Unity Technologies	2024			False	False	True	https://unity.com/resources/ultimate-guide-advanced-multiplayer-networking
7	Unity Gaming Services	Unity Technologies	Unity Technologies				False	False	True	https://docs.unity.com/ugs/en-us/manual/overview/manual/unity-gaming-services-home

8 LO(s)

No.	CLO Name	CLO Details
1	CLO1	Understand SOLID principles and design patterns in game development.
2	CLO2	Ability to use common Unity design patterns, such as Singleton, Observer, Command, Strategy, Factory, Decorator, and State Pattern.
3	CLO3	Understanding profiling in Unity
4	CLO4	Understand how to optimize game performance.
5	CLO5	Understand the Unity Netcode solution and Unity Services.
6	CLO6	The ability to utilize Unity AI and apply AI models to game projects is essential.
7	CLO7	Develop multiplayer game applications using Unity.
8	CLO8	Collaborate effectively in a team environment for game development projects.

View mapping of CLOs to PLOs

Download All Student Material

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Stu Tas
1	Course Introduction & Topic Allocation.	Offline		IT		AGU301	
2	Course Introduction & Topic Allocation.	Offline		IT		AGU301	
3	Chapter 1: SOLID and Common Design Pattern in Game Development 1.1 The SOLID principles	Offline	CLL01, CLO2	T	https://unity.com/resources/design-patterns-solid-ebook		- Re slid rela refe doc - Do exe - Se prac
4	Chapter 1: SOLID and Common Design Pattern in Game Development 1.1 The SOLID principles	Offline	CLL01, CLO2	T	https://unity.com/resources/design-patterns-solid-ebook		- Re slid rela refe doc - Do exe - Se prac
5	Chapter 1: SOLID and Common Design Pattern in Game Development 1.1 The SOLID principles	Offline	CLL01, CLO2	T	https://unity.com/resources/design-patterns-solid-ebook		- Re slid rela refe doc - Do exe - Se prac
6	Chapter 1: SOLID and Common Design Pattern in Game Development 1.1 The SOLID principles	Offline	CLL01, CLO2	T	https://unity.com/resources/design-patterns-solid-ebook		- Re slid rela refe doc - Do exe - Se prac
7	Chapter 1: SOLID and Common Design Pattern in Game Development 1.2 Design patterns for game development	Offline	CLL01, CLO2	T	https://unity.com/resources/design-patterns-solid-ebook		- Re slid rela refe doc - Do exe - Se prac

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Stu Tas
8	Chapter 1: SOLID and Common Design Pattern in Game Development 1.2 Design patterns for game development	Offline	CLL01, CLO2	T	https://unity.com/resources/design-patterns-solid-ebook		- Re slid rela refe doc - Do exe - Se prac
9	Chapter 1: SOLID and Common Design Pattern in Game Development 1.2 Design patterns for game development	Offline	CLL01, CLO2	T	https://unity.com/resources/design-patterns-solid-ebook		- Re slid rela refe doc - Do exe - Se prac
10	Chapter 1: SOLID and Common Design Pattern in Game Development 1.2 Design patterns for game development	Offline	CLL01, CLO2	T	https://unity.com/resources/design-patterns-solid-ebook		- Re slid rela refe doc - Do exe - Se prac
11	Chapter 1: SOLID and Common Design Pattern in Game Development 1.2 Design patterns for game development	Offline	CLL01, CLO2	T	https://unity.com/resources/design-patterns-solid-ebook		- Re slid rela refe doc - Do exe - Se prac
12	Chapter 1: SOLID and Common Design Pattern in Game Development 1.2 Design patterns for game development	Offline	CLL01, CLO2	T	https://unity.com/resources/design-patterns-solid-ebook		- Re slid rela refe doc - Do exe - Se prac
13	Assessment 1 (Week3): Milestone 1 Review & Evaluation	Offline	CLO1, CLO2	TU	Presentation slides, report		Pres mile rece feed
14	Assessment 1 (Week3): Milestone 1 Review & Evaluation	Offline	CLO1, CLO2	TU	Presentation slides, report		Pres mile rece feed

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Stu Tas
15	Quiz 1	Offline	CLO1, CLO2	U	Quiz 1		Do
16	Chapter 2: Optimize your game performance 2.1 Profiling	Offline	CLO3	T	https://unity.com/resources/optimize-your-console-and-pc-game-performance		- Re slid rela refe doc - Do exe - Se prac
17	Chapter 2: Optimize your game performance 2.1 Profiling	Offline	CLO3	T	https://unity.com/resources/optimize-your-console-and-pc-game-performance		- Re slid rela refe doc - Do exe - Se prac
18	Chapter 2: Optimize your game performance 2.1 Profiling	Offline	CLO3	T	https://unity.com/resources/optimize-your-console-and-pc-game-performance		- Re slid rela refe doc - Do exe - Se prac
19	Chapter 2: Optimize your game performance 2.2 Use the Memory Profiler	Offline	CLO4	T	https://unity.com/resources/optimize-your-console-and-pc-game-performance		- Re slid rela refe doc - Do exe - Se prac
20	Chapter 2: Optimize your game performance 2.2 Use the Memory Profiler	Offline	CLO4	T	https://unity.com/resources/optimize-your-console-and-pc-game-performance		- Re slid rela refe doc - Do exe - Se prac
21	Chapter 2: Optimize your game performance 2.2 Use the Memory Profiler	Offline	CLO4	T	https://unity.com/resources/optimize-your-console-and-pc-game-performance		- Re slid rela refe doc - Do exe - Se prac

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Stu Tas
22	Chapter 2: Optimize your game performance 2.3 Programming and code architecture	Offline	CLO4	T	https://unity.com/resources/optimize-your-console-and-pc-game-performance		- Re slid rela refe doc - Do exe - Se prac
23	Chapter 2: Optimize your game performance 2.3 Programming and code architecture	Offline	CLO4	T	https://unity.com/resources/optimize-your-console-and-pc-game-performance		- Re slid rela refe doc - Do exe - Se prac
24	Chapter 2: Optimize your game performance 2.3 Programming and code architecture	Offline	CLO4	T	https://unity.com/resources/optimize-your-console-and-pc-game-performance		- Re slid rela refe doc - Do exe - Se prac
25	Chapter 2: Optimize your game performance 2.4 Unity profiling and debug tools	Offline	CLO4	T	https://unity.com/resources/optimize-your-console-and-pc-game-performance		- Re slid rela refe doc - Do exe - Se prac
26	Chapter 2: Optimize your game performance 2.4 Unity profiling and debug tools	Offline	CLO4	T	https://unity.com/resources/optimize-your-console-and-pc-game-performance		- Re slid rela refe doc - Do exe - Se prac
27	Quiz 2	Offline	CLO3, CLO4	U	Quiz 2		Do
28	Chapter 3: Unity networking solutions 3.1 Netcode for GameObjects and Netcode for Entities 3.2 Setting up your first Netcode project. Network synchronization	Offline	CLO5, CLO7	T	https://docs.unity3d.com/Packages/com.unity.netcode@1.8/manual/index.html		- Re slid rela refe doc - Do exe - Se prac

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Stu Tas
29	Chapter 3: Unity networking solutions 3.1 Netcode for GameObjects and Netcode for Entities 3.2 Setting up your first Netcode project. Network synchronization	Offline	CLO5, CLO7	T	https://docs.unity3d.com/Packages/com.unity.netcode@1.8/manual/index.html		- Re slid rela refe doc - Do exe - Se prac
30	Chapter 3: Unity networking solutions 3.1 Netcode for GameObjects and Netcode for Entities 3.2 Setting up your first Netcode project. Network synchronization	Offline	CLO5, CLO7	T	https://docs.unity3d.com/Packages/com.unity.netcode@1.8/manual/index.html		- Re slid rela refe doc - Do exe - Se prac
31	Chapter 4: Third-party networking	Offline	CLO7	TU	Critical thinking, AI literacy		- Se prac
32	Chapter 4: Third-party networking	Offline	CLO7	TU	Critical thinking, AI literacy		- Se prac
33	Chapter 4: Third-party networking	Offline	CLO7	TU	Critical thinking, AI literacy		- Se prac
34	Assessment 2 (Week 6):Milestone 2 Review & Evaluation	Offline	CLO3, CLO4, CLO7	TU	Presentation slides, report		Pres mile rece feed
35	Assessment 2 (Week 6):Milestone 2 Review & Evaluation	Offline	CLO3, CLO4, CLO7	TU	Presentation slides, report		Pres mile rece feed
36	Assessment 2 (Week 6):Milestone 2 Review & Evaluation	Offline	CLO3, CLO4, CLO7	TU	Presentation slides, report		Pres mile rece feed

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Stu Tas
37	Chapter 5: Inference Engine (Unity Sentis) 5.1 Understand the Inference Engine workflow 5.2 Models in Inference Engine	Offline	CLO6	TU	https://docs.unity3d.com/Packages/com.unity.ai.inference@2.3/manual/index.html		- Re slid rela refe doc - Do exe - Se prac
38	Chapter 5: Inference Engine (Unity Sentis) 5.1 Understand the Inference Engine workflow 5.2 Models in Inference Engine	Offline	CLO6	TU	https://docs.unity3d.com/Packages/com.unity.ai.inference@2.3/manual/index.html		- Re slid rela refe doc - Do exe - Se prac
39	Chapter 5: Inference Engine (Unity Sentis) 5.1 Understand the Inference Engine workflow 5.2 Models in Inference Engine	Offline	CLO6	TU	https://docs.unity3d.com/Packages/com.unity.ai.inference@2.3/manual/index.html		- Re slid rela refe doc - Do exe - Se prac
40	Chapter 5: Inference Engine (Unity Sentis) 5.3 Use tensors	Offline	CLO6	TU	https://docs.unity3d.com/Packages/com.unity.ai.inference@2.3/manual/index.html		- Re slid rela refe doc - Do exe - Se prac
41	Chapter 5: Inference Engine (Unity Sentis) 5.3 Use tensors	Offline	CLO6	TU	https://docs.unity3d.com/Packages/com.unity.ai.inference@2.3/manual/index.html		- Re slid rela refe doc - Do exe - Se prac
42	Chapter 5: Inference Engine (Unity Sentis) 5.3 Use tensors	Offline	CLO6	TU	https://docs.unity3d.com/Packages/com.unity.ai.inference@2.3/manual/index.html		- Re slid rela refe doc - Do exe - Se prac

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Stu Tas
43	Chapter 6: AI Assistant	Offline	CLO6, CLO7	TU	https://docs.unity3d.com/Packages/com.unity.ai.assistant@1.0/manual/index.html		- Re slid rela refe doc - Do exe - Se prac
44	Chapter 6: AI Assistant	Offline	CLO6, CLO7	TU	https://docs.unity3d.com/Packages/com.unity.ai.assistant@1.0/manual/index.html		- Re slid rela refe doc - Do exe - Se prac
45	Chapter 6: AI Assistant	Offline	CLO6, CLO7	TU	https://docs.unity3d.com/Packages/com.unity.ai.assistant@1.0/manual/index.html		- Re slid rela refe doc - Do exe - Se prac
46	Assessment 3 (Week 8):Milestone 3 Review & Evaluation	Offline	CLO5, CLO6, CLO7	TU	Presentation slides, report		Pres mile rece feed
47	Assessment 3 (Week 8):Milestone 3 Review & Evaluation	Offline	CLO5, CLO6, CLO7	TU	Presentation slides, report		Pres mile rece feed
48	Quiz 3	Offline	CLO5, CLO6	U	Quiz 3		Do
49	Chapter 7: AI generator 7.1 Sprite Generator and Texture2D Generator 7.2 Sound Generator	Offline	CLO6, CLO7	TU	https://docs.unity3d.com/Packages/com.unity.ai.generators@latest		- Re slid rela refe doc - Do exe - Se prac
50	Chapter 7: AI generator 7.1 Sprite Generator and Texture2D Generator 7.2 Sound Generator	Offline	CLO6, CLO7	TU	https://docs.unity3d.com/Packages/com.unity.ai.generators@latest		- Re slid rela refe doc - Do exe - Se prac

Session	Topic	Learning-Teaching Type	LO	ITU	Student Materials	S-Download	Stu Tas
51	Chapter 7: AI generator 7.1 Sprite Generator and Texture2D Generator 7.2 Sound Generator	Offline	CLO6, CLO7	TU	https://docs.unity3d.com/Packages/com.unity.ai.generators@latest		- Re slid rela refe doc - Do exe - Se prac
52	Chapter 7: AI generator 7.3 Animation Generator & Material Generator	Offline	CLO6, CLO7	TU	https://docs.unity3d.com/Packages/com.unity.ai.generators@latest		- Re slid rela refe doc - Do exe - Se prac
53	Chapter 7: AI generator 7.3 Animation Generator & Material Generator	Offline	CLO6, CLO7	TU	https://docs.unity3d.com/Packages/com.unity.ai.generators@latest		- Re slid rela refe doc - Do exe - Se prac
54	Chapter 7: AI generator 7.3 Animation Generator & Material Generator	Offline	CLO6, CLO7	TU	https://docs.unity3d.com/Packages/com.unity.ai.generators@latest		- Re slid rela refe doc - Do exe - Se prac
55	Final Project Presentation	Offline	CLO7, CLO8	U	Presentation slides, report		Pres mile rece feed
56	Final Project Presentation	Offline	CLO7, CLO8	U	Presentation slides, report		Pres mile rece feed
57	Final Project Presentation	Offline	CLO7, CLO8	U	Presentation slides, report		Pres mile rece feed
58	Final Project Presentation	Offline	CLO7, CLO8	U	Presentation slides, report		Pres mile rece feed
59	Final Project Presentation	Offline	CLO7, CLO8	U	Presentation slides, report		Pres mile rece feed
60	Final Project Presentation	Offline	CLO7, CLO8	U	Presentation slides, report		Pres mile rece feed

0 Constructive question(s)

4 assessment(s)

No.	Category	Type	Part	Weight	Completion Criteria	Duration	CLO	Question Type	No Question	Knowledge and Skill	Grading Guide	Note
1	Assessment (PBL milestones check)	on-going	3	30.0%	>0	20'/group	Assessment 1 (Week3): CLO1, CLO2 Assessment 2 (Week 6): CLO3, CLO4, CLO7 Assessment 3 (Week 8): CLO5, CLO6, CLO7	The student presents slides, gives a demonstration, and answers questions from the lecturer.	By instructor	Assessment 1: CLO1, CLO2 Assessment 2: CLO3, CLO4, CLO7 Assessment 3: CLO5, CLO6, CLO7	in class, by instructor	Instructor has responsibility to review the test for students after graded.
2	Final Project Presentation	on-going	1	30.0%	5	30'/group	CLO7, CLO8	The student presents slides, gives a demonstration, and answers questions from the lecturer.	By instructor		in class, by instructor	Instructor has responsibility to review the test for students after graded.
3	Progress test (Quiz)	on-going	3	15.0%	>0	30'	Quiz 1: Chapter 1; CLO1, CLO2 Quiz 2: Chapter 2; CLO3, CLO4 Quiz 3: Chapter 4, 5, 6; CLO5, CLO6	Multiple choices Marked by Computer or a suitable format	25	Quiz 1 cover content: Chapter 1, 2 Quiz 2 cover content: Chapter 3, 4 Quiz 3 cover content: Chapter 5, 6, 7	by exam board, using computer	Instruction and schedules for Progress tests must be presented in the Course Implementation Plan approved by director of the campus. Progress test must be taken right after the last lectures of required material. Instructor has responsibility to review the test for students after graded.
4	Final Exam	Final exam	1	25.0%	4	60'	CL1 - CLO6	Multiple choices Marked by Computer	50	concepts, algorithms; all studied chapters	by exam board, using computer	The exam questions must be updated or different at least 70% to the previous ones.